

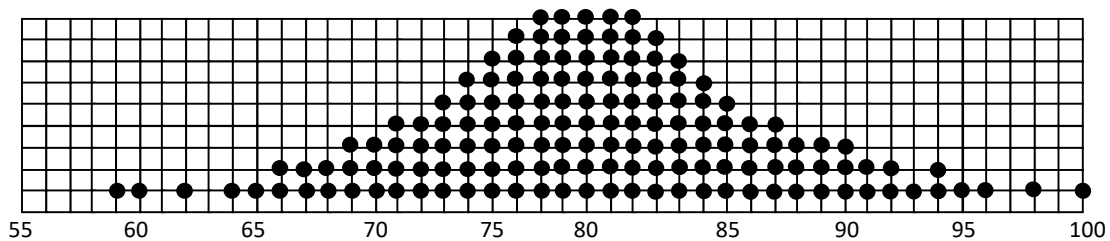
Agenda

- Selected slides from this week
 - More traditional description
- Time permitting: Grading, Evaluation, Academic Honesty, Deadline extension, “what if I missed”, ...

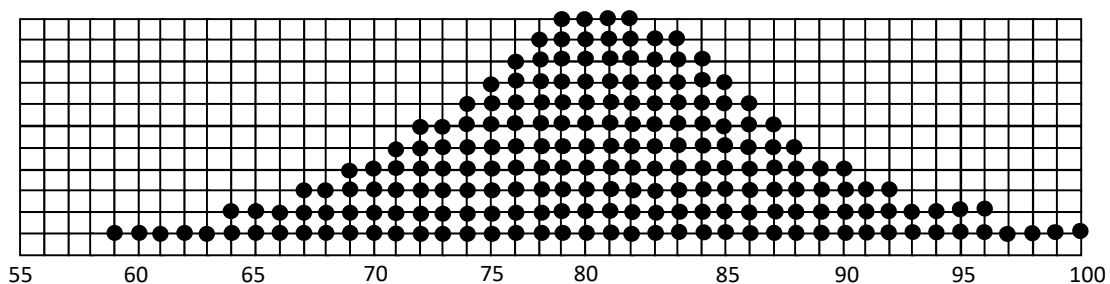
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Pulse Rate Plot

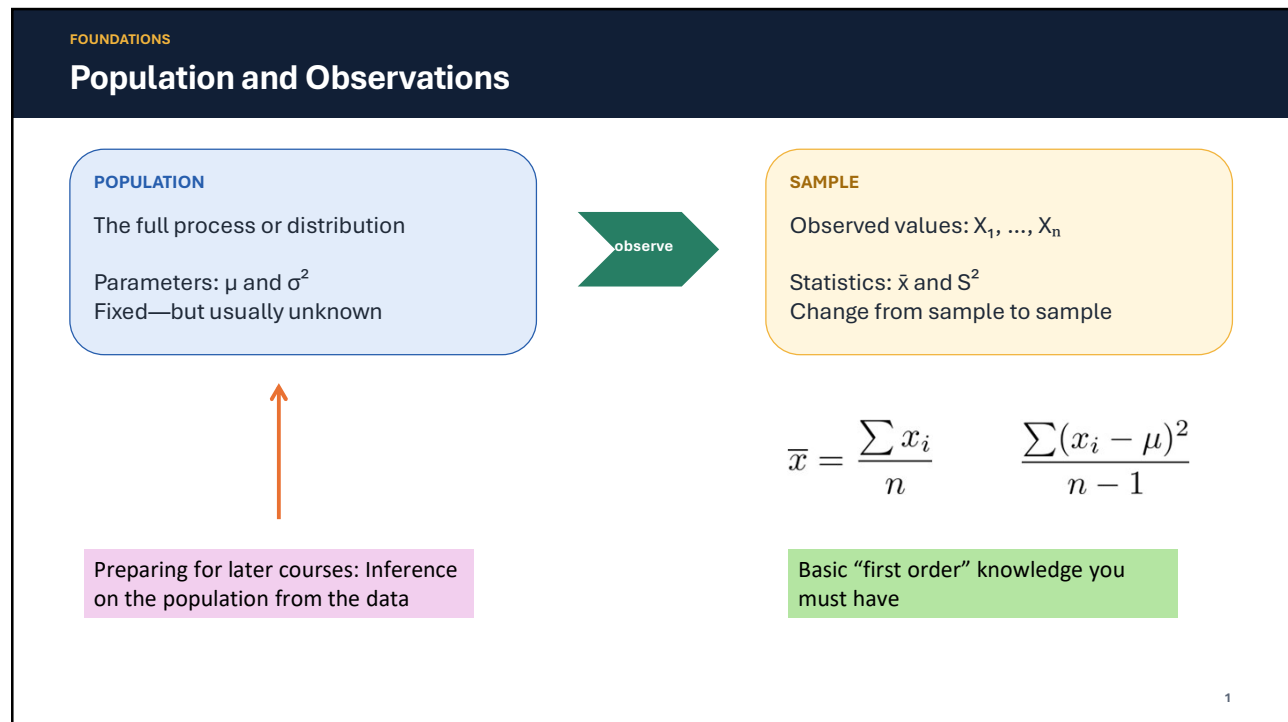
The distribution approaches a symmetrical distribution



200 students



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Probability Space

- A set S of all possible outcomes of a random experiment
- Event: A subset of the sample space (“an observation”)
- Example:
 - An experiment whose outcome is not known in advance: A coin toss
 - We do know the complete set of possible outcomes – in this case Heads or Tails.
 - If $E = \{H\}$, then E is the event that the coin produced a heads.

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Probability Space

- A set S of all possible outcomes of a random experiment
- Event: A subset of the sample space (“an observation”)
- Probability:
 - A non-negative number associated with an event X
 - For any sequence of mutually exclusive events $X_1, X_2, \dots$ and the union Y , $P(Y) = \text{sum of individual probabilities}$
 - $P(S) = 1$
- Space: The triple $\{S, E, P\}$
 - P is often written as $\Pr$

Pro Tip: Mutually Exclusive Events $\neq$ Independent Events

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Random Variables

- In many random experiments we care not about the observed values themselves, but about some numerical quantity
 - Example: the sum of two dice throws, or the number of heads in n consecutive coin tosses
- Any such quantity determined by the result of a random experiment is a *random variable*
 - Technically: A random variable is a function defined on a probability space $\{S, E, P\}$
 - $X: S \rightarrow V$ where V is a set of numerical values, e.g., set of integers or real numbers
- Random variables can be discrete or continuous

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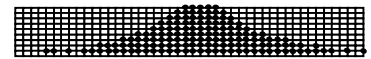
Distribution function



- A random variable is ~~synonymous~~ with a cumulative distribution function, or simply a distribution function
 - We define the cumulative distribution function (cdf) $F_X(x) = \Pr\{X \leq x\}$ (x is real)
 - Specifying the cdf is like implicitly defining the random variable
- Notation: $X \sim F_X$
- All probability questions about X can be answered in terms of F
 - Example to follow

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Discrete Random Variables



- Random variables whose values form a finite or countably infinite sequence are discrete random variables.
 - Example: the outcome of a coin toss or a dice throw.
- The probability mass function $p_X(x) = P(X = x)$
- Example: Sum of two dice throws

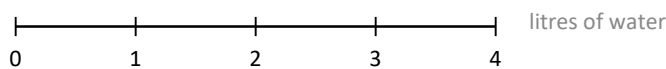
x	2	3	4	5	6	7	8	9	10	11	12
$P(X=x)$	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

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Continuous Random Variable

- Random variables that take values across a continuum are continuous random variables.
 - Ex: Amount of water in a 4-litre jar: continuous random variable on $[0, 4]$
- Unlike a discrete random variable, the probability of any **particular** value in the continuum is **zero**.
 - There are infinitely many values in $[0, 4]$, and each is equally likely.

Basic “first order” concept



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Continuous Random Variable

- *Formally*, X is a continuous random variable if there exists a non-negative function $f(x)$ defined for all real x having the property that for any set C of real numbers

$$\Pr\{X \in C\} = \int_{x \in C} f(x) dx$$

Pro Tip: This seemingly roundabout way is to get over the “difficulty” mentioned in the previous slide

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Continuous random variable: Example

Consider a cdf of the form

$$F_X(x) = \begin{cases} 0, & x \leq 0 \\ 1 - e^{-x^2}, & x > 0 \end{cases}$$

The probability that X exceeds 1

$$P(X > 1) = 1 - P(X \leq 1) = 1 - F_X(1) = e^{-1}$$

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Continuous Random Variable

- *Formally*, X is a continuous random variable if there exists a non-negative function $f(x)$ defined for all real x having the property that for any set C of real numbers

$$\Pr\{X \in C\} = \int_{x \in C} f(x) dx$$

- The function $f(x)$ is called the probability density function (pdf)

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Continuous Random Variable

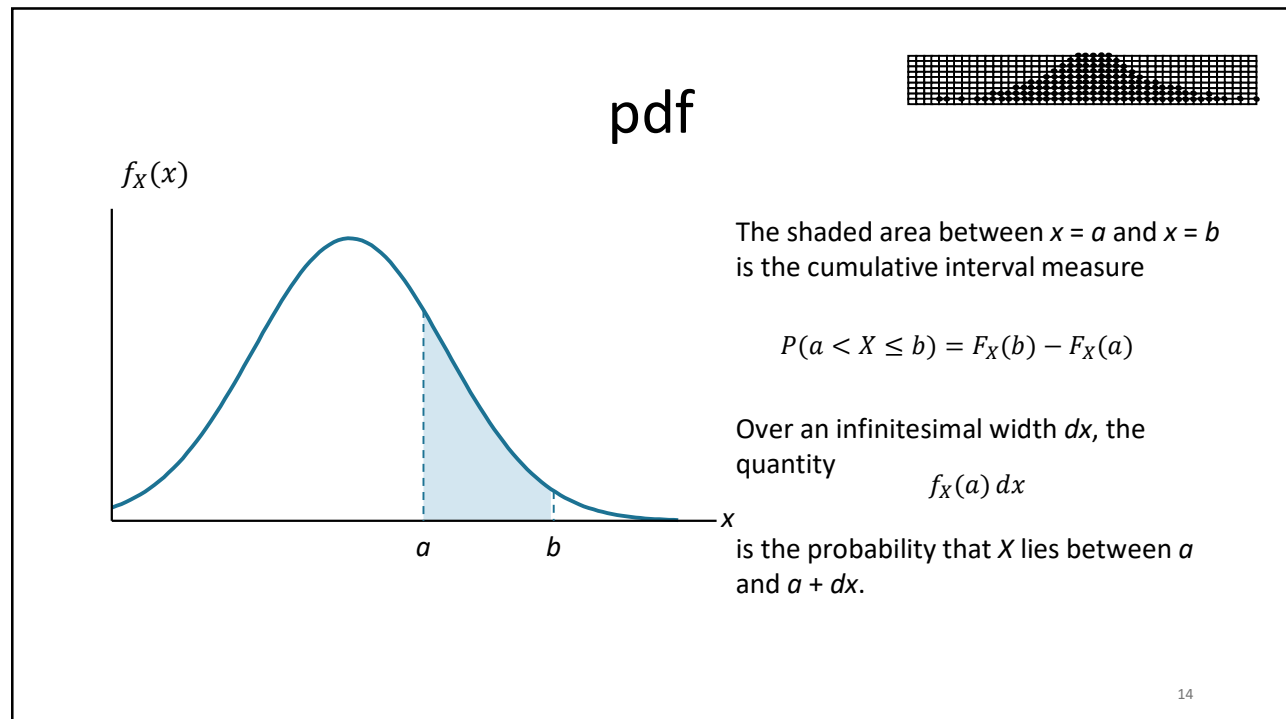
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$$\Pr\{X \in C\} = \int_{x \in C} f(x) dx$$

- The function $f(x)$ is called the probability density function (pdf)
- The relationship between f and F :

$$F(a) = \Pr\{X \in (-\infty, a]\} = \int_{-\infty}^a f(x) dx$$

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Probability density function (pdf)

The pdf of X at x is the **derivative** of its cdf at x .

$$f_X(x) = \frac{d}{dx} F_X(x)$$

It is a non-negative function such that for any set B of real numbers

$$P\{X \in B\} = \int_B f_X(x) dx$$

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Probability Density

- The probability density function (pdf) is the derivative of the cumulative distribution function (cdf) at that value x

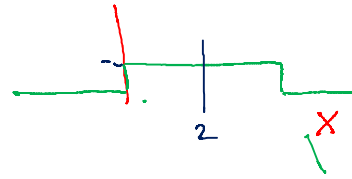
$$f_X(a) = \lim_{\epsilon \rightarrow 0} \frac{\Pr\left\{a - \frac{\epsilon}{2} \leq X \leq a + \frac{\epsilon}{2}\right\}}{\epsilon}$$

$$\Pr\left\{a - \frac{\epsilon}{2} \leq X \leq a + \frac{\epsilon}{2}\right\} \approx \epsilon f(a) = \int_{a-\frac{\epsilon}{2}}^{a+\frac{\epsilon}{2}} f_X(x) dx$$

Pro Tip: Notice how we get probability of a particular value=0 as desired

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Expectation



For a **discrete** random variable X :

$$E(X) = \sum_i x_i P(X = x_i)$$

For a **continuous** random variable X :

$$E(X) = \int_{-\infty}^{+\infty} x f_X(x) dx$$

Expected value of X is also called *mean* and represented by the symbol μ

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Expected value of a function of a random variable

For a function $g(X)$ of a **discrete** random variable $E(g(X)) = \sum_i g(x_i) P(X = x_i)$

For a **continuous** random variable $E(g(X)) = \int_{-\infty}^{+\infty} g(x) f_X(x) dx$

Expected value of $(X - \mu)^2$ is called *variance* and represented by the symbol σ^2

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FOUNDATIONS

Population and variance

POPULATION

The full process or distribution

Parameters: μ and σ^2
Fixed—but usually unknown

- Suppose someone (an “oracle”) told you that values are
 - Only ($X = 0$ or $X = 2$ or $X = 4$)
 - What’s the mean and variance?
 - Expected value: $\mu = 2$
 - Variance: $\sigma^2 = 8/3$

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TWO DIFFERENT CALCULATIONS

Oracle knowledge vs sample estimation

QUESTION	ORACLE: POPULATION KNOWN	STATISTICIAN: ONLY A SAMPLE
Information	$P(X=0)=P(X=2)=P(X=4)=1/3$	Observed sample: $X_1, \dots, X_n$
Center used	True mean $\mu = 2$	Sample mean $\bar{x}$
Quantity	$\sigma^2 = E[(X-\mu)^2]$	$S^2 = \sum (X_i - \bar{x})^2 / (n-1)$
Nature	A fixed population parameter	A random estimator; it varies
Result here	$\sigma^2 = (4+0+4)/3 = 8/3$ exactly	$E[S^2] = 8/3$ across repeated samples

Same number, different roles: 8/3 is the oracle target and the estimator long-run average.

POPULATION → SAMPLE → INFERENCE

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